

RQ Answers

omitted

2024-04-05

Data overview

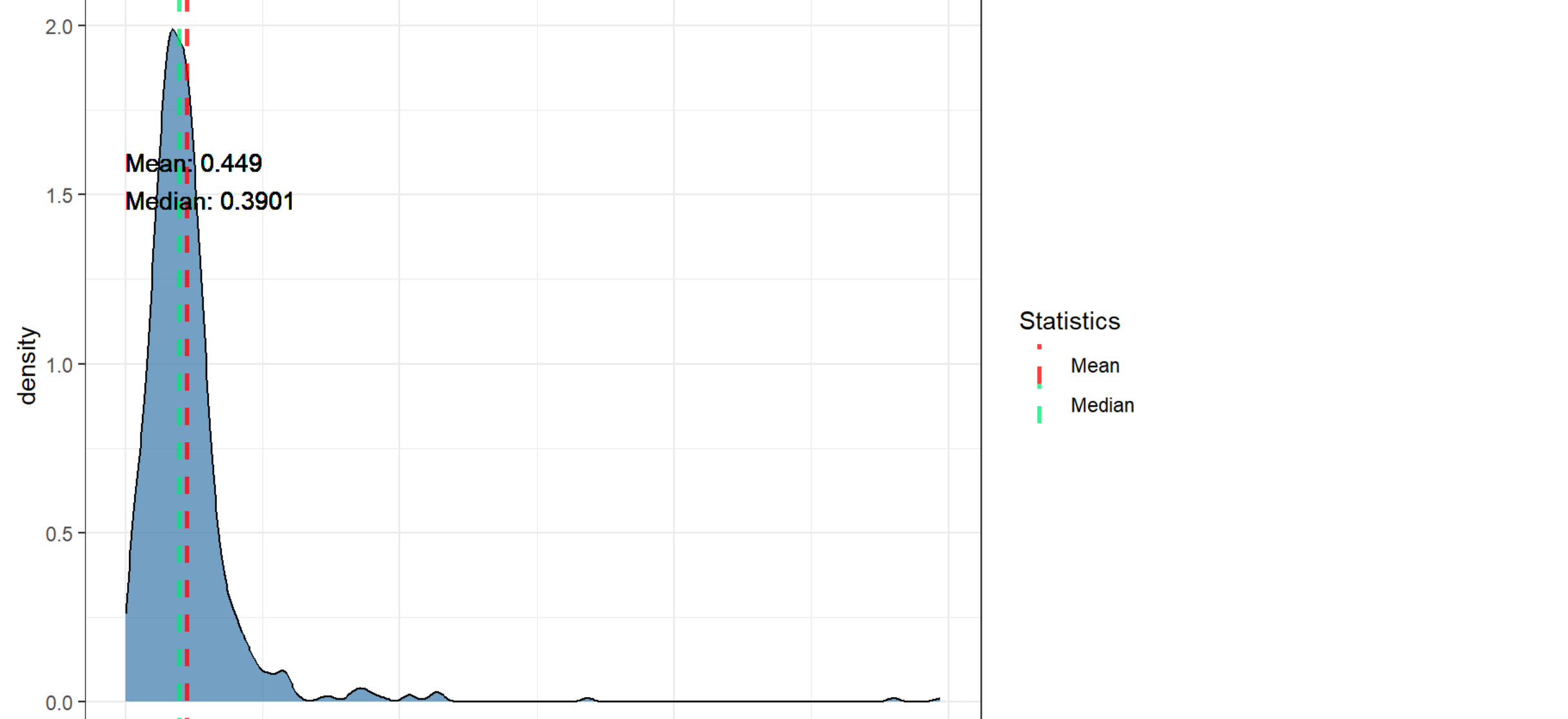
glimpse(all_data)				
## Rows: 131,133				
## Columns: 11				
## \$ repo		<chr>	"accumulo", "accumulo", "accumulo", "accumulo", "a...	
## \$ pr_number		<int>	1302, 1302, 1302, 1302, 1302, 1302, 1302, 13...	
## \$ rule		<chr>	"java:S1192", "java:S3776", "java:S1319", "java:S1...	
## \$ severity		<chr>	"CRITICAL", "CRITICAL", "MINOR", "CRITICAL", "CRIT...	
## \$ file		<chr>	"core/snc/main/java/org/apache/accumulo/core/metad...	
## \$ type		<chr>	"CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...	
## \$ status		<chr>	"OPEN", "OPEN", "OPEN", "CLOSED", "OPEN", "...	
## \$ debt		<int>	8, 6, 10, 5, 6, 2, 5, 15, 30, 109, 30, 30, 5, 5, 5...	
## \$ ncloc_affected_file		<int>	347, 347, 347, 347, 347, 347, 347, 347, 252, 252, ...	
## \$ complexity		<int>	68, 68, 68, 68, 68, 68, 68, 68, 86, 86, 86, 86...	
## \$ origin		<chr>	"NEW", "NEW", "NEW", "NEW", "PRE-EXISTING", "PRE-E...	

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the data has 771 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but the data has 771 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```

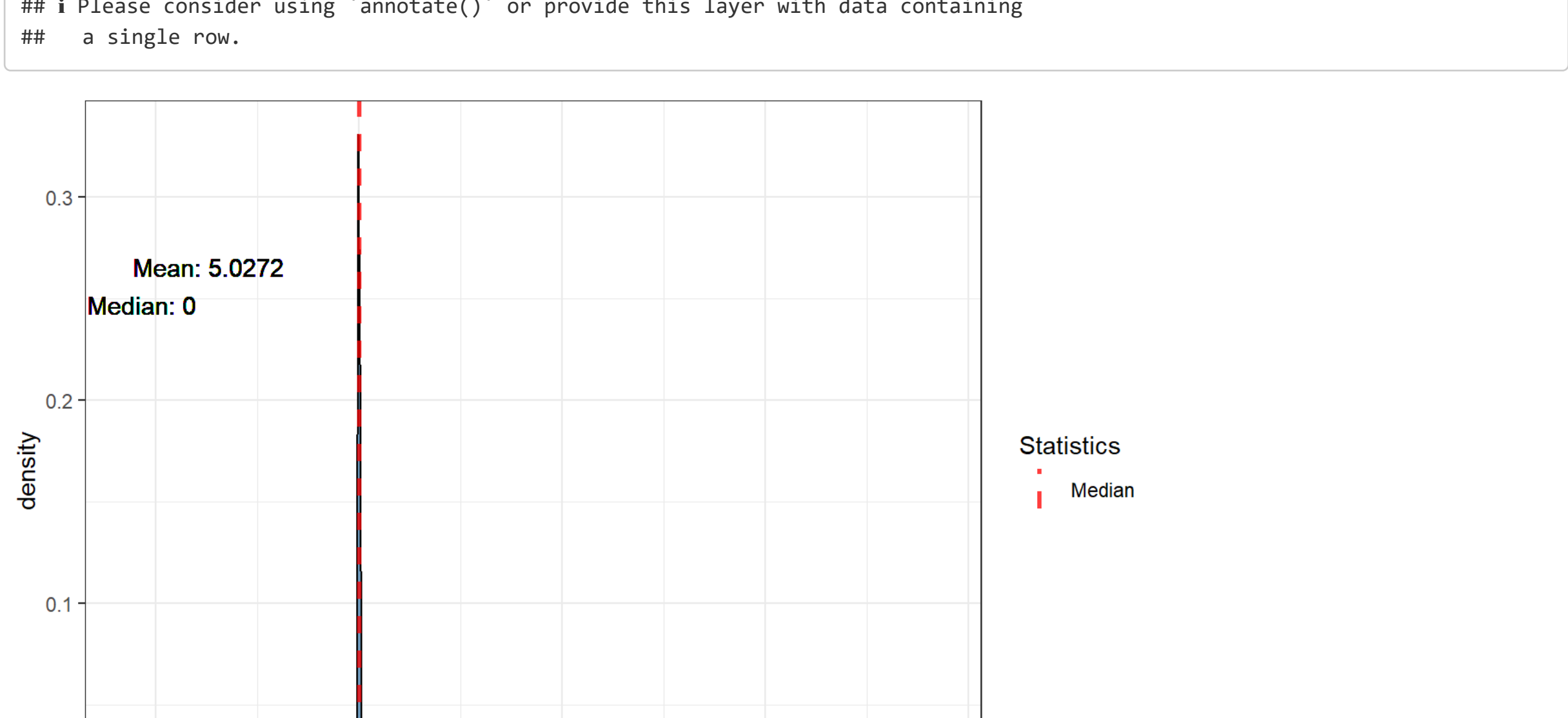


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the data has 771 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but the data has 771 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	
## 1 accumulo	26.5	53.4	20.1	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3			
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>	<dbl>
## 1	26.5	53.4	20.1

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3			
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 accumulo	62.7	37.3	

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	62.7	37.3

Percentages for new TDV by repo

## # A tibble: 1 × 3			
## # Groups: repo [1]			
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 accumulo	73.2	26.8	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	73.2	26.8

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank accumulo		
## <int> <chr>		
## 1	1 java:S117	- 593
## 2	2 java:S2589	- 367
## 3	3 java:S112	- 364
## 4	4 java:S1125	- 259
## 5	5 java:S101	- 246
## 6	6 java:S2293	- 228
## 7	7 java:S3776	- 225
## 8	8 java:S1155	- 193
## 9	9 java:S1192	- 155
## 10	10 java:S100	- 137

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S117	122	
## 2 java:S2589	123	
## 3 java:S112	124	
## 4 java:S1125	125	
## 5 java:S101	126	
## 6 java:S2293	127	
## 7 java:S3776	128	
## 8 java:S1155	129	
## 9 java:S1192	130	
## 10 java:S100	131	

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank accumulo		
## <int> <chr>		
## 1	1 java:S117	- 16750
## 2	2 java:S2589	- 15126
## 3	3 java:S101	- 10684
## 4	4 java:S1125	- 9873
## 5	5 java:S3776	- 7882
## 6	6 java:S112	- 6346
## 7	7 java:S1192	- 6245
## 8	8 java:S2293	- 6111
## 9	9 java:S116	- 4711
## 10	10 java:S2142	- 4656

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S117	122	
## 2 java:S2589	123	
## 3 java:S101	124	
## 4 java:S1125	125	
## 5 java:S3776	126	
## 6 java:S112	127	
## 7 java:S1192	128	
## 8 java:S2293	129	
## 9 java:S116	130	
## 10 java:S2142	131	

Outliers

Higher TDD

## # A tibble: 771 × 5				
## repo	pr_number	total_debt	total_ncloc	tdd
## <chr>	<int>	<int>	<int>	<dbl>
## 1 accumulo	2922	410	69	5.94
## 2 accumulo	4215	140	25	5.6
## 3 accumulo	3612	724	215	3.37
## 4 accumulo	2591	333	145	2.30
## 5 accumulo	2865	374	165	2.27
## 6 accumulo	2907	383	171	2.24
## 7 accumulo	2597	307	148	2.07
## 8 accumulo	2297	120	58	2.07
## 9 accumulo	3540	30	16	1.88
## 10 accumulo	2180	250	139	1.80
## # i 761 more rows				

Higher TDV

## # A tibble: 771 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	2022 accumulo	14300	
## 2	1946 accumulo	3875	
## 3	2335 accumulo	935	
## 4	1891 accumulo	643	
## 5	1605 accumulo	508	
## 6	3827 accumulo	422	
## 7	3563 accumulo	392	
## 8	3756 accumulo	371	
## 9	1905 accumulo	337	
## # i 761 more rows			

Lower TDV

## # A tibble: 771 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	3604 accumulo	-5740	
## 2	3402 accumulo	-4826	
## 3	3229 accumulo	-724	
## 4	2705 accumulo	-724	
## 5	3117 accumulo	-691	
## 6	2966 accumulo	-600	
## 7	3327 accumulo	-508	
## 8	3890 accumulo	-506	
## 9	2156 accumulo	-405	
## 10	2168 accumulo	-396	
## # i 761 more rows			

Extra

Number of PRs with pre-existing TD

## n	
## 1 766	

Number of java:S1192 issues that affect test code

## n	
## 1 0	

Number of java:S1192 issues that affect not test code

## n	
## 1 6400	

Summary NCLOC

## ncloc	
## Min. :	15
## 1st Qu.:	267
## Median :	711
## Mean :	3187
## 3rd Qu.:	1735
## Max. :	202678